A Comparison of Orbit Determination Algorithms for a Low Earth Orbit Satellite

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2. Abstract

This report presents the development, implementation and evaluation of a MATLAB-based orbit determination algorithm for a low Earth orbit satellite, which is modeled after NASA's QuickSCAT mission. The satellite's orbit is a near-circular solar-synchronous orbit at an altitude of 800km and an inclination of 98.6° and is tracked and observed by three ground stations that provide range and range-range data over a five hour tracking period. The linearized observation model relating the 18-element state vector composed of: satellite position, velocity, parameters, drag coefficient, station coordinates is incorporated with a nonlinear force model that includes two-body gravity, J_2 oblateness , and exponential atmospheric drag. The two estimation algorithms that are implemented and compared are the Batch least-squares regression model and the conventional sequential Kalman filter with the Joseph covariance formulation. The batch processor converged to post-fit residuals of 0.0097 m in range and 0.000998 m/s in range-rate with a position error ellipsoid semi axes on the order of millimeters. The sequential Kalman filter produced post-fit residuals of 0.1161 m in range and 0.001405 m/s in range-rate with a position error ellipsoid around one order of magnitude larger than the batch estimator. The comparison of post-fit residuals confirm the absence of global optimality that is linked with the sequential estimator. Additionally, to evaluate a more theoretically accurate version of determining the covariance matrix, the Joseph formulation was implemented and compared to the CKF's approach in determining $P0$. Multiple trade studies were conducted including ground station network geometry, a priori covariance sensitivity, and data type strength analysis all confirmed that range and range-rate observations are geometrically complementary and that using both in conjunction is necessary to obtaining a well-conditioned orbit determination solution. The batch and sequential estimating algorithms are essential tools to solve orbit determination problems in different contexts from post-processing to real-time navigation applications respectively.

3. The Orbit Determination Problem

Accurate orbit determination of satellites is a fundamental problem in spacecraft guidance and navigations with vast applications ranging from Earth observation and telecommunications to scientific endeavors. Orbit determination requires fusing tracking data with dynamic models to produce the most accurate possible estimate of a spacecraft's state from imperfect data of initial conditions, force models, and measurement systems.

The reference tracking data is in the form of range and range-rate observations collected by three distinct ground stations: a ship in the Pacific Ocean assumed to be fixed (Station 101), a ground station in Pirinclik, Turkey (Station 337), and a ground station in Thule, Greenland (Station 394). Simulated measurement noise with standard deviation of 1 cm and 1 mm/s were added to the range and range-rate observations respectively over the 5 hours through which the data was measured. Station 101 is assumed to be fixed and treated as a reference due to the impracticability of modelling ship motion and its initial covariance is set to roughly zero.

Two different estimation algorithms are implemented and compared. The first one is a batch least-squares processor, which accumulates all observations simultaneously to solve for the state correction needed by minimizing the weighted sum of the squared residuals. The second algorithm is the conventional Kalman filter, which processes observations one at a time, using the state transition matrix to propagate the reference trajectory and the error covariance between measurement epochs. To maintain numerical stability, the Joseph formulation is also considered. In all cases, the dynamics are linearized about a reference trajectory.

4. Theory

4.1 Batch Least Squares Estimation

Batch least squares estimation relies on the idea that all orbit measurements are available before the orbit determination problem begins. It relies on the assumption of linearized dynamics about a reference trajectory, and can be iterated 2-3 times to minimize linearization error during the approximation. Batch estimation can be performed without an *a priori* estimate, but for the case of this project an *a priori* estimate is considered.

From the system dynamics in Section 3 and the STM at the initial epoch, the equations of motion can be integrated through a 5th order Runge-Kutta to obtain the mapping between the satellite state at the initial epoch to the desired time. Then, the Jacobian of the observation model G can be utilized to determine the sensitivity of a measurement at the desired time relative to the satellite state at the initial epoch. Following this, the information matrix λ and the normal matrix *N* are updated and can be utilized to find the estimated state covariance matrix and the estimated state correction.

Additionally, the measurement noise covariance matrix R provides significant impacts to each observation's contribution to λ and *N*. A smaller variance in R for a specific measurement type results in the filter trusting that observation type more.

Notably however, the Batch Processor struggles when inverting a poorly scaled matrix, and is additionally subject to a range of numerical issues when the resulting covariance matrix becomes not symmetric and therefore non-positive definite.

4.2 Sequential Estimation (CKF)

The sequential estimation algorithm continually updates the estimates of the spacecraft state and uncertainty as new measurements are recorded. The traditional sequential estimation algorithm is subject to numeric issues and is computationally expensive as it requires the inversion of a $n\ x\ n$ matrix to compute the covariance and estimated state deviation, where n is the number of state variables considered (18). As a result, the traditional sequential processing algorithm is manipulated to yield a computationally less expensive filter, called the conventional Kalman filter, which requires the inversion of only a $p \ x \ p$ matrix, where p is the number of measurements for the current timestep (2).

The conventional Kalman filter performs a 'Time Update' and a 'Measurement Update' for each observation recorded. The time update utilizes the STM determined similarly in Section 4.1 to update the *a priori* state and covariance estimation at the current timestep before considering the new measurement. The measurement update then weighs the contribution of the new observation through the Kalman Gain magnitude to the best estimate of the state deviation and the best estimate of the covariance, reducing uncertainty and leading to filter convergence.

4.3 Joseph Covariance Formulation

The Joseph Formulation has several main advantages in comparison to the traditional formulation of the covariance matrix P seen in the conventional Kalman Filter. The traditional formulation for P assumes that the computed K is exactly the optimal Kalman gain, when in reality small numerical issues can lead to P becoming asymmetric or no longer positive definite. The Joseph Formulation enforces a symmetric result by construction regardless of the numerical value of K, but it does not assure P to be positive definite. This still results in a substantial improvement in filter convergence as it prevents the increasingly large uncertainties leading to a degradation of kalman gain and finally filter divergence.

5. Software Architecture and Implementation

Our solution to the orbit-determination problem for the specified mission is made up of a set of MATLAB scripts. The main algorithms are main.m, *project_batch.m*, *project_sequential.m*, and *project_joseph.m* and all call the *project_setup.m* script to load the observation data, initialize the state vector and covariance, and define the physical constants for the problem. The dynamics of the algorithm are handled by the *project_dyn.m*, the observation partials are done through the *computed_Htilde.m* script, and the STM partials are computed in the *computed_A.m* . This modular approach ensures that both the all implemented filters depend on the same inputs and models, enabling easy and direct comparison between the chosen methods.

6. Discussion & Analysis

6.1 Observation Station Analysis

6.1.1 Rationale for Fixing Station 101

The Pacific Ocean ship sensor - Station 101 - was chosen to be a fixed position in the initial problem setup by defining its covariance in the filter to be approximately zero. We chose to model this particular station as fixed since we do not have approximate dynamics models, like wind, desired navigation course, ocean current, signal interference and more to approximately model the ships movement properly across the ocean. As such, if we considered Station 101 to not be a fixed station and did not implement any additional dynamics models, the estimated position of Station 101 would likely oscillate and its covariance would likely blow up over time.

6.1.2 The Effect of Omitting a Station from the Solution

If we did not solve for an estimation of the true position of one of the stations in the ECI frame while considering its position to not be fixed, we would obtain a slightly different estimate of the true state $\hat{x}$ since the other 15 state variables must account for the unmodeled bias induced by utilizing an inaccurate station position. Similarly, we would obtain a covariance matrix P with greater 1-sigma uncertainty for the satellite state variables since the filter is unable to correlate a portion of the measurement uncertainty to the unsolved for station coordinates. This can be visualized in Figures 6.1.1a and 6.1.1b below.

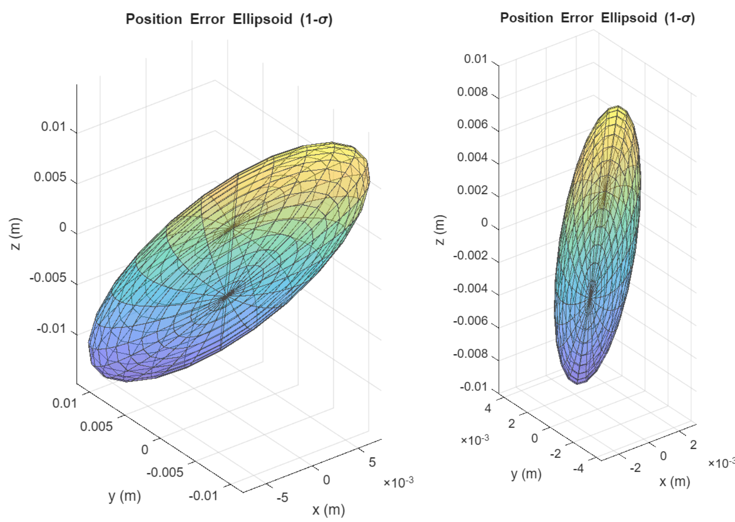

Figure 6.1.1: (a, left) Position-error ellipsoid for the batch state-error covariance matrix. (b, right) Position-error ellipsoid for the batch state-error covariance matrix without solving for station 394's ECI coordinates.

6.1.3 Sensitivity to the Choice of Fixed Station

Yes, fixing one station instead of another creates different impacts to the estimated true spacecraft state vector. Although the two non-fixed stations share the same *a priori* covariance of 10^6, they are at significantly different locations across Earth, meaning that their observation data contains different information content. Each station has significantly different observability of the spacecraft's trajectory due to their inherent relative geometry. Additionally, a singular station can not determine the spacecraft's motion in all three dimensions since the station can only measure the distance and rate of change of distance along its line of sight. This can be observed in Figures 6.1.2a and 6.1.2b below.

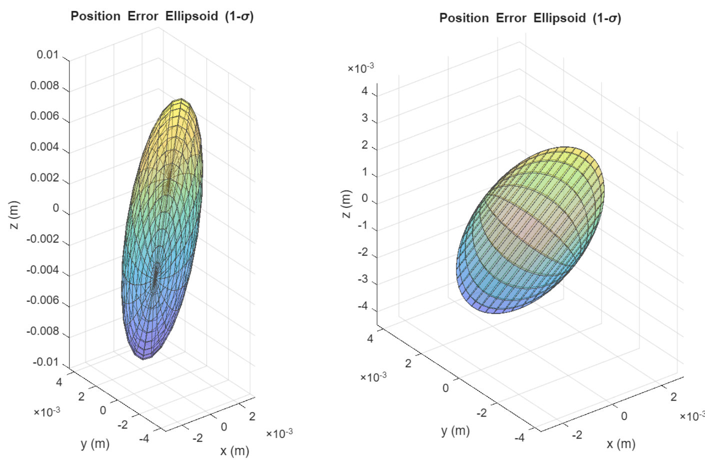

Figure 6.1.2: (a, left) Batch Position-error ellipsoid omitting station 394's coordinates. (b, right) Batch Position-error ellipsoid omitting station 337's coordinates.

6.2 Impacts of A Priori Covariance and Data-noise

6.2.1 Effect of A Priori Covariance

The *a priori* covariance and measurement-noise covariance both impact the results of the batch filter during the computation of the information matrix lambda.

The *a priori* covariance $\bar{P}_0$ tells the selected algorithm how confident it should be in the *a priori* state $\bar{x}_0$. As such, by increasing the magnitude of the entries of the *a priori* covariance, the estimator has more freedom to explore more of the design space instead of staying local to the initial guess. A small magnitude of the entries of the *a priori* covariance means that we are extremely confident in the initial state estimate and as a result the filter will avoid deviating from that guess.

6.2.2 Effect of Measurement Noise Covariance

The measurement-noise covariance R indicates how much the filter is able to trust the measured observation from each of the tracking stations. The values of R can be affected by randomized signal interference, miscalibrations, onboard delays, etc. Therefore, large magnitudes of the entries in R tells the filter to rely less on the measured observation from the station, instead relying on the *a priori* data. Small magnitudes of the entries of R on the other hand tell the filter that the observations are highly accurate and will therefore decrease the post-fit residuals.

6.2.3 Effect of Tighter A Priori

If we used an *a priori* more compatible with the actual errors in the initial conditions, then we would rely more heavily on the *a priori* state instead of the station observations during the computation of lambda. As a result, the measurements from the stations would have a marginal impact on the estimated values of the true spacecraft state, as the information matrix would be dominated from the contribution of $\overline{P}_0^{-1}$.

6.3 Batch Algorithm Pre-fit and Post-fit residuals

The batch algorithm pre-fit and post-fit residuals can be seen in Table 6.3.1 and Figures 6.3.1a and 6.3.1b below.

Table 6.3.1: Pre-fit and Post-fit residuals resulting from the Batch Least Squares Algorithm

	ρ (m)	$\frac{d\rho}{dt}$ (m/s)
Pre-fit RMS (first pass)	732.7483	2.900165
Post-fit RMS	0.0097	0.000998

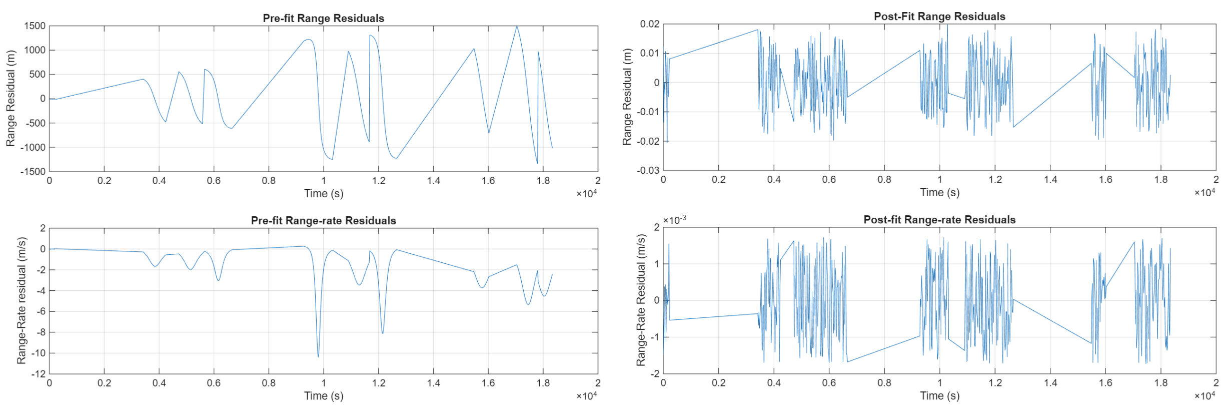

Figure 6.3.1: The Batch Algorithm (a) Pre-fit and (b) Post-fit residuals for range and range-rate.

6.4 Batch Algorithm Position-error ellipsoid

The position-error ellipsoid (Figure 6.4.1) for the Batch Algorithm was obtained through an eigendecomposition of the final covariance matrix $P0$. The position covariance entries (1:3,1:3) were extracted from P0, and then the determined eigenvectors V define the principal axis of the ellipsoid while the eigenvalues D indicate the variance in each direction. The standard deviation was obtained by taking the square root of D. As a result, the position-error ellipsoid showcases the direction and magnitude of the resulting uncertainty in the x, y, and z ECI frame orientation.

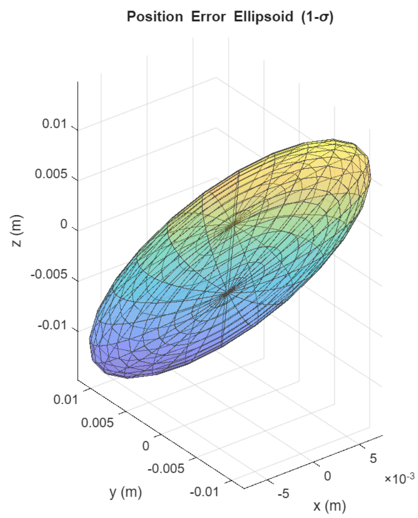

Figure 6.4.1: Position-error ellipsoid for the batch state-error covariance matrix.

6.5 Investigation into relative strengths of the range and range-rate data

To investigate the relative contributions of the range and range-rate data in the Batch algorithm, individual solutions omitting the non-desired data were obtained. This was done by collapsing vectors y, H, $H^{\sim}$, and W to only account for the corresponding observation data.

The pre-fit and post-fit residuals can be seen in Table 6.5.1 and Figure 6.5.1 below.

Table 6.5.1: Pre and Post-fit residuals for the Batch Algorithm with a measurement type omitted.

	Only ρ (m)	Only $\frac{d\rho}{dt}$ (m/s)
Pre-fit RMS (first pass)	732.7483	2.900165
Post-fit RMS	0.0097	0.000978

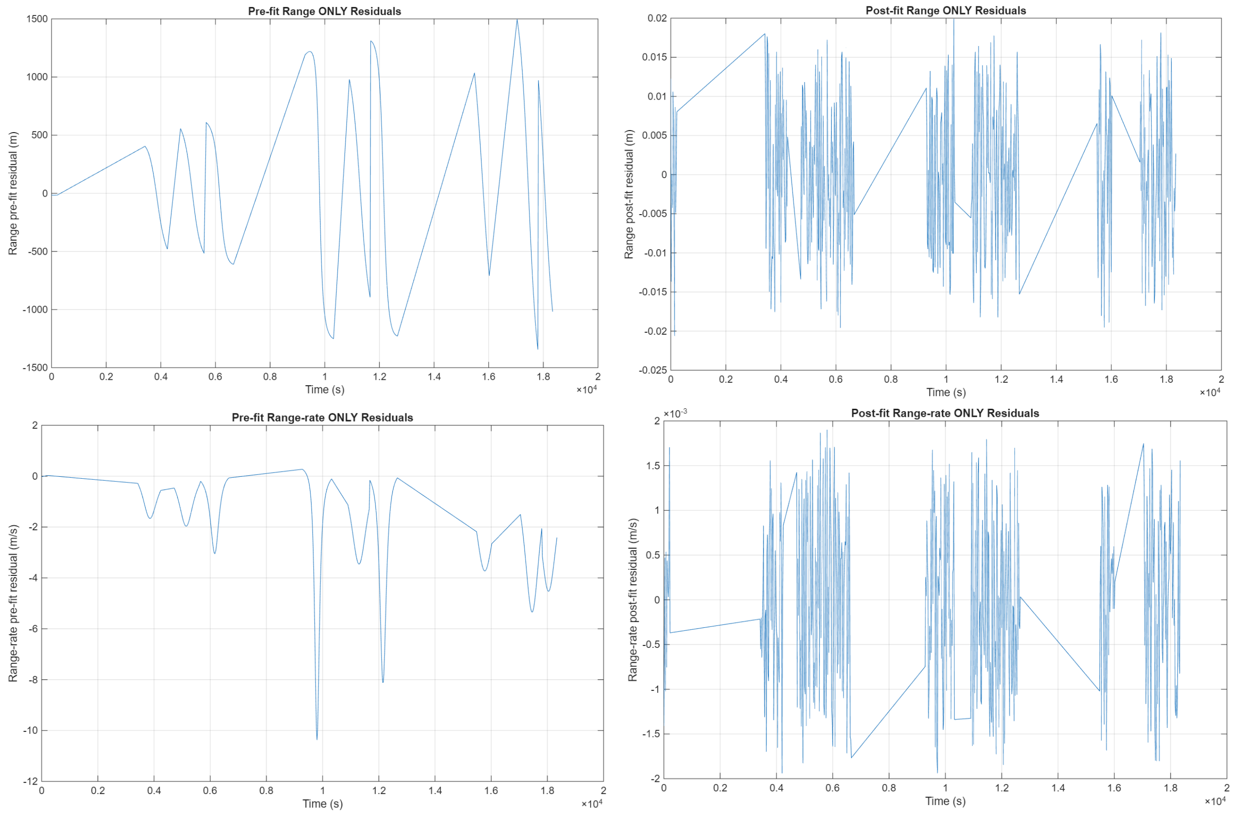

Figure 6.5.1: Batch Algorithm (left) Pre-fit and (right) Post-fit residuals for the [TOP] Range and [BOTTOM] Range-rate only scenarios.

Both the range-only and range-rate-only solutions produce the same results for the pre-fit RMS values of 732.7483 m and 2.900165 m/s respectively. The post-fit RMS values of 0.0097 m and 0.000978 m/s are nearly identical to each other. This outcome is expected as the pre-fit residuals are computed before any correction is applied and as such they reflect only the initial reference trajectory's error. Meaningful comparison between the two filters thus requires not examining the residuals, but in comparing the geometry of the respective position error ellipsoids.

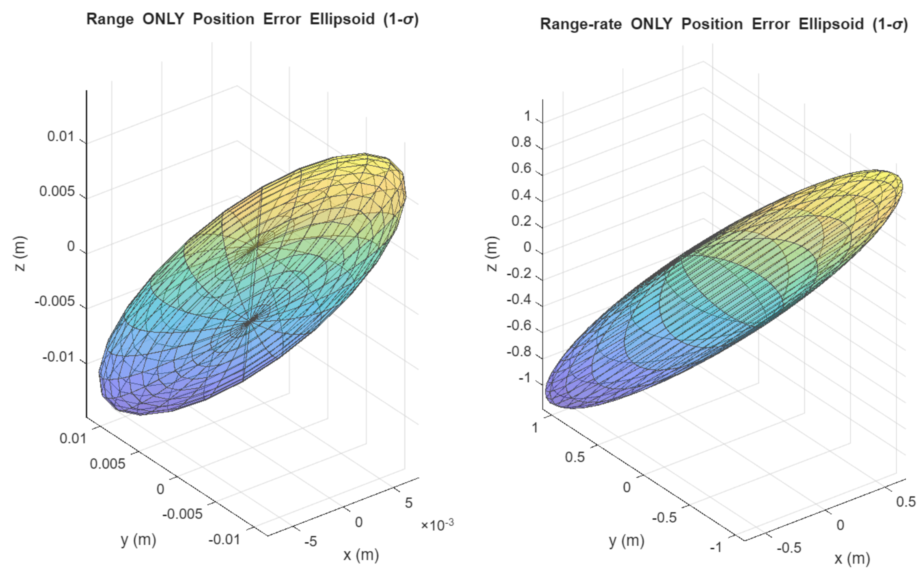

Figure 6.5.2: The position-error ellipsoids for the (left) Range and (right) Range-rate scenarios.

However, the position-error ellipsoids shown in Figure 6.5.2 show a distinct difference in the physical geometry of the uncertainty. The ellipsoid describing the range-only position error is flat with oblate geometry, which reflects high sensitivity to radial position, motion toward or away from the station, but lower sensitivity to motion perpendicular to the line of sight, corresponding to the along-track and cross-track directions.

On the other hand, the range-rate-only ellipsoid is much larger in volume in all three ECI coordinates. This makes sense as range-rate measures the derivative of distance, meaning distance has to be calculated indirectly by integrating velocity information over time. Calculating velocity indirectly leads to a larger accumulated error resulting in much larger position uncertainty. Range measurements, by directly measuring the distance from a known station location, yield absolute position results more accurately, which creates a far smaller error ellipsoid.

Overall, range data contributes more to absolute position accuracy while producing a compact and well-constrained error ellipsoid. Range-rate-only data on the other hand, creates significantly more uncertainty in absolute position due to solving indirectly for position by integrating velocity measurements. Therefore, using a combination of both types of data is essential for a well-rounded solution to the orbit determination problem.

6.6 Implementation of a Kalman Filter

The pre and post-fit residuals of range and range-rate resulting from the conventional Kalman filter can be seen in Table 6.6.1 and Figure 6.6.1 below.

Table 6.6.1: Pre-fit and Post-fit residuals resulting from the conventional Kalman filter.

	ρ (m)	$\frac{d\rho}{dt}$ (m/s)
Pre-fit RMS	732.1844	0.579943
Post-fit RMS	0.1161	0.001405

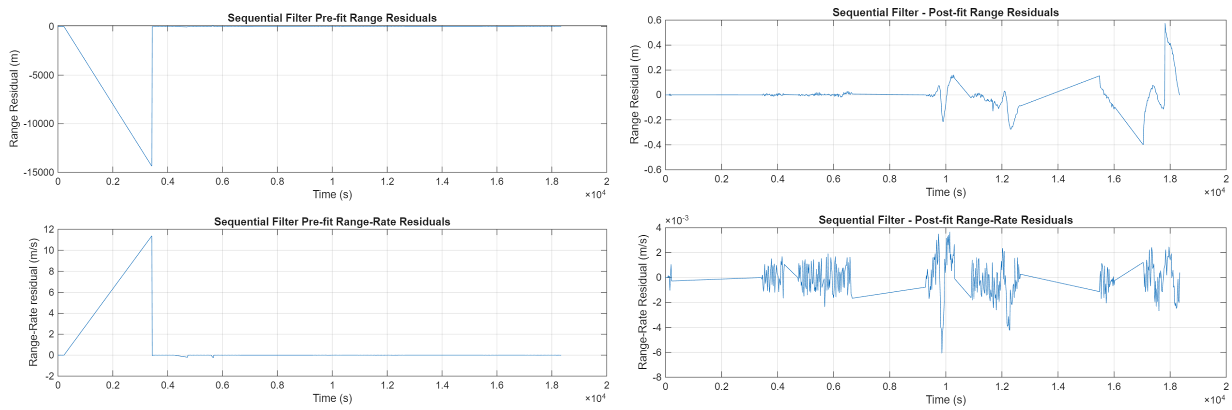

Figure 6.6.1: Sequential Kalman filter (a) Pre and (b) Post-fit residuals for range and range-rate.

In comparison to the Batch algorithm, the post-fit RMS of the conventional Kalman filter is greater for both range and range-rate. This is because not only does the Batch algorithm complete multiple passes, it also has the key advantage of obtaining a globally optimal solution which minimizes the residual across all observations simultaneously. In other words, the inherent nature of the conventional Kalman filter as a sequential processor restricts it from analyzing more than one observation at a time to update the state estimate. This tradeoff in global optimality is exchanged for the Kalman filter's ability to process new measurements in real time while the Batch algorithm must recompute its solution across all n observations, which may grow to be computationally expensive.

Additionally, the post-fit residual plot for range-rate in Figure 6.6.1 for the conventional Kalman filter showcases the potential issues that arise when there are large gaps between measurement data. The spikes and volatility exhibited are due to the ~3000 second gaps in the observation data between stations. This behavior is evident in the Batch algorithm RMS as well, but to a significantly lesser extent since the Batch algorithm again minimizes the residual across all observations globally while the Kalman filter accumulates error since it has no future knowledge of what the next set of observations might look like.

Plots of the covariance position and velocity trace shown in Figure 6.6.2 below.

Figure 6.6.2: Conventional Kalman filter (top) position and (bottom) velocity covariance trace.

While no numerical stability issues showed up in the output of the conventional Kalman filter, numerical issues can be addressed by implementing the Joseph formulation for covariance or by enforcing the covariance matrix *P0* to be symmetric through $P\_i = (P\_i + P\_i') / 2$. Notably however, when both were employed as seen in Figure 6.6.3, numerical issues (seen as the discontinuities in both the position and velocity covariance were apparent.

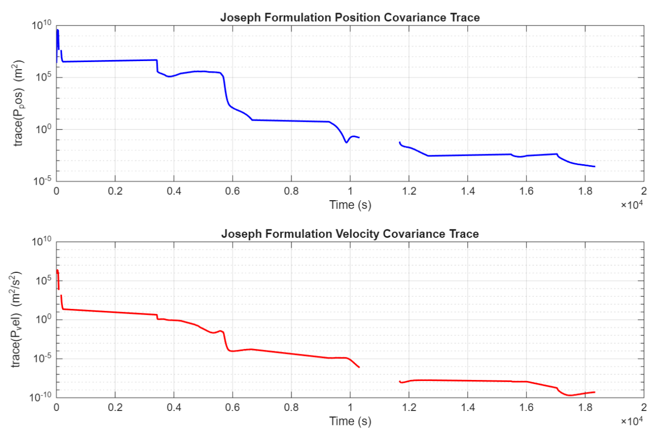

Figure 6.6.3: Joseph Formulation (top) position and (bottom) velocity covariance trace exhibiting instabilities shown as the discontinuities in the plots.

While this is an unexpected result, this can likely be attributed to the large gaps in the observation data, which leads to severe ill-conditioning which is unexposed by the less accurate formulation for covariance found in the conventional Kalman filter.

One additional significant difference between the Sequential filter and the traditional Batch least squares filter is that the linearization errors accumulate over time in the Sequential filter since the system is not reiterating and relinearizing about its new *a priori* state. The Batch filter does this through taking multiple passes, driving the estimated true state of the satellite towards its true state and minimizing the state uncertainty. While not implemented as part of this project, the extension of the conventional Kalman filter - the extended Kalman filter (EKF) - relinearizes about the *a priori* state as each observation is processed, approaching the globally optimal solution achieved by the Batch filter.

7. Conclusion

This report presents the development and validation of an orbit determination algorithm for a low Earth orbit satellite modeled after NASA's QuickSCAT mission. Two distinct orbit determination estimation algorithms are implemented and compared. The first algorithm is a Batch least squares-regression model. The second one is a conventional sequential Kalman filter with the Joseph covariance form used to tackle the numerical issues associated with the covariance calculation in the conventional kalman filter.

The batch processor's post-fit residuals converged to 0.0097 m in range and 0.000998 m/s in range-rate with a position error ellipsoid semi axes on the order of millimeters. Additionally, it was found that due to the inherent geometry between the observation stations and the satellite, the choice of which station is fixed has significant impacts on the uncertainty in satellite position as well as the best estimate of the satellite's true state. Further, the relationship between the magnitude of the *a priori* covariance as well as the measurement noise covariance was investigated, revealing that both have impacts on the relative weighting between each other during the computation of the information matrix lambda.

The data type strength analysis showed that range and range-rate observables can complement each other as range provides strong sensitivity to radial position and range-rate provides high sensitivity to along-track velocity. The combined use of both data types yields a significantly better conditioned position solution when compared to using just one of the two observables in isolation.

The sequential Kalman filter converged to post-fit residuals of 0.1161 m in range and 0.001405 m/s in range-rate with a position error ellipsoid around one order of magnitude larger than the batch estimator. A numerical instability made apparent in the conventional covariance update was resolved using the Joseph formulation, which enforces a symmetric result. The covariance trace plots confirmed a key advantage of the sequential Kalman filter, which is its ability to process new measurements in real time with a notable disadvantage being related to the Kalman filter being a sequential processor, which restricts it from analyzing more than one observation at a time to update the state estimate.

In conclusion, both the sequential Kalman filter and the Batch estimator represent two methods suitable for different situations in orbit determination. The batch estimator is best used in post-processing when all observation data is available and high accuracy is necessary. The sequential Kalman filter is best used for real-time navigation purposes such as for onboard spacecraft navigation where observation data must be processed continuously as it arrives.

References

[1] Rosengren, Aaron. MAE 182: Lecture Slides